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DATA
SET

DS1406

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.⁹ Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

Educational Level	Married		
	Yes	No	
Bachelors or higher	187	113	300
Some college	162	138	300
High school or less	149	151	300
	498	402	900

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

H_0 : marriage rate and educational level are independent

H_a : marriage rate and educational level are dependent

$$\chi^2 = \frac{(O_i - E_i)^2}{E_i} = \frac{(187-166)^2}{166} + \frac{(162-166)^2}{166} + \frac{(149-166)^2}{166} + \frac{(113-134)^2}{134} + \frac{(138-134)^2}{134} + \frac{(151-134)^2}{134} \approx 10.0611 > \chi^2_{2,0.01} = 9.210$$

$\Rightarrow$ Reject H_0 , 在 $\alpha = 0.01$ 下, 結婚率與不同教育程度之間存在顯著的差異

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

Family Members	Type of Residence			
	Apartment	Duplex	Single Residence	
1	8	20	1	29
2	16	8	9	33
3	10	10	14	34
4 or More	6	2	16	24
	40	40	40	120

Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

H_0 : 三種居住類型的家庭人口數分佈相同

H_a : 三種居住類型的家庭人口數分佈不同

$$\chi^2 = \frac{(8-9.67)^2}{9.67} + \frac{(20-9.67)^2}{9.67} + \frac{(1-9.67)^2}{9.67} + \frac{(16-11)^2}{11} + \frac{(8-11)^2}{11} + \frac{(9-11)^2}{11} \\ + \frac{(10-11.33)^2}{11.33} + \frac{(10-11.33)^2}{11.33} + \frac{(14-11.33)^2}{11.33} + \frac{(6-8)^2}{8} + \frac{(2-8)^2}{8} + \frac{(16-8)^2}{8}$$

$$= 36.499 > \chi^2_{6,0.01} = 16.812 \Rightarrow \text{Reject } H_0$$

這三種居住類型的家庭人口數分配存在顯著差異

Duplex 傾向於單身 & 小家庭居住, single residence 傾向人口較多的大家庭居住, Apartment 人數分佈介於兩者之間

12. Dissolved Oxygen Content Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

Above Town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

- a. Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.
- b. Use a Student's t -test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

$$\bar{X}_A = 5, \quad S_A^2 = 0.025$$

$$\bar{X}_B = 4.86, \quad S_B^2 = 0.013$$

$$S_p^2 = \frac{4 \times 0.025 + 4 \times 0.013}{5 + 5 - 2} = 0.019$$

a.
 H_0 : 上游、下游溶氧量分佈相同

H_a : 上游溶氧量大於下游的溶氧量

$$T_1 = 34, \quad T_1^* = 5(5+5+1) - 34 = 21, \quad T = \min(34, 21) = 21$$

$$\alpha = 0.05, \quad n_1 = 5, \quad n_2 = 5 \Rightarrow \text{查表得臨界值為 } 19, \quad T = 21 > 19$$

$\Rightarrow$ Non-Reject $H_0 \Rightarrow$ 在 $\alpha = 0.05$ 下沒有足夠證據顯示該城鎮排放污水導致溶氧量顯著下降

b.
 $H_0: \mu_1 - \mu_2 \leq 0, \quad H_a: \mu_1 - \mu_2 > 0$

$$t = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{S_p^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{5 - 4.86}{\sqrt{0.019 \left(\frac{1}{5} + \frac{1}{5} \right)}} = 1.606 < t_{8, 0.05} = 1.860$$

$\Rightarrow$ Non-Reject H_0 , 結論同(a)

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

	Deaf Children	Hearing Children
	2.75 (15)	.89 (1)
	2.14 (11)	1.43 (7)
	3.23 (18)	1.06 (4)
	2.07 (10)	1.01 (3)
	2.49 (14)	.94 (2)
	2.18 (12)	1.79 (8)
	3.16 (17)	1.12 (5.5)
	2.93 (16)	2.01 (9)
	2.20 (13)	1.12 (5.5)
Rank Sum	126	45

H_0 : 兩母體的 eye-movement rates 分布相同

H_a : 兩母體分布不同

$$T_1 = 126, T_1^* = 9(9+9+1) - 126 = 45, T = \min(126, 45) = 45$$

$$n_1 = 9, n_2 = 9, \alpha = 0.05 \text{ 臨界值為 } b_2, T = 45 \leq b_2 \Rightarrow \text{Reject } H_0$$

在 $\alpha = 0.05$ 下，有充分證據顯示聾啞兒童和聽力正常兒童的 eye-movement rates 分布不同

Two Simple Examples Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \leq .10$. Calculate the observed value of the test statistic and present your conclusion.

4.

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
	+	+	+	-	+	+	+

$$\Rightarrow X = 6$$

H_0 : 兩母體分布相同

H_a : 兩母體分布不同

$$\alpha \leq 0.1$$

$$X \stackrel{H_0}{\sim} \text{Bin}(1, 0.5)$$

$$P(X=0) = \binom{1}{0} 0.5^1 = 0.0018$$

$$P(X=1) = \binom{1}{1} 0.5^1 = 0.0541$$

$$P(X \leq 1) + P(X \geq 6) = 2 \times (0.0018 + 0.0541) = 0.1118 > 0.1$$

$$P(X=0) + P(X=1) = 2 \times 0.0018 = 0.0036 < 0.1$$

$\Rightarrow$ R.R. $X=0$ or $X=1 \Rightarrow X=6$ 不落入 R.R.

$\Rightarrow$ Non-Reject H_0 #

11. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1–10. The data are shown in the table. Do the data provide sufficient evidence to indicate that one of the chefs tends to give higher ratings than the other? Test by using the sign test with a value of α near .05.

Meal	A	B		Meal	A	B	
1	6	8	-	12	8	5	+
2	4	5	-	13	4	2	+
3	7	4	+	14	3	3	x
4	8	7	+	15	6	8	-
5	2	3	-	16	9	10	-
6	7	4	+	17	9	8	+
7	9	9	x	18	4	6	-
8	7	8	-	19	4	3	+
9	2	5	-	20	5	4	+
10	4	3	+	21	3	2	+
11	6	9	-	22	5	3	+

$$+ = 11$$

$$- = 9$$

- Use the binomial tables in Appendix I to find the exact rejection region for the test.
- Use the large-sample z statistic. (NOTE: Although the large-sample approximation is suggested for $n \geq 25$, it works fairly well for values of n as small as 15.)
- Compare the results of parts a and b.

$$H_0: p = 0.5, \quad H_a: p \neq 0.5$$

$$a. \quad X \stackrel{H_0}{\sim} \text{Bin}(20, 0.5)$$

$$P(X \leq 4) = 0.0059, \quad \underbrace{P(X \leq 5) = 0.0201}_{\downarrow \text{接近 } \alpha = 0.025}, \quad P(X \leq 6) = 0.0571$$

$$\Rightarrow R.R. = X \leq 5 \text{ or } X \geq 15, \quad X = 11 \text{ 不落入 } R.R.$$

$$\Rightarrow \text{Non-reject } H_0$$

$$b. \quad z = \frac{X - \mu}{\sigma} = \frac{X - np}{\sqrt{np(1-p)}} = \frac{11 - 20 \times 0.5}{\sqrt{20 \times 0.5 \times 0.5}} = 0.447 < z_{0.05} = 1.96$$

$$\Rightarrow \text{Non-reject } H_0$$

c. (a), (b) 的方法皆得到相同結論

$\Rightarrow$ 沒有足夠的證據顯示兩位廚師中有一位給出比另一位更高的評分

13. Recovery Rates Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs.

- Test using the sign test. Choose your rejection region so that α is near .05.
- Why might it be inappropriate to use the Student's t -test in analyzing the data?

Hospital	Drug A		
	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	84	63	75.0
2	63	44	69.8
3	56	48	85.7
4	77	57	74.0
5	29	20	69.0
6	48	40	83.3
7	61	42	68.9
8	45	35	77.8
9	79	57	72.2
10	62	48	77.4

Hospital	Drug B		
	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	96	82	85.4
2	83	69	83.1
3	91	73	80.2
4	47	35	74.5
5	60	42	70.0
6	27	22	81.5
7	69	52	75.4
8	72	57	79.2
9	89	76	85.4
10	46	37	80.4

$n = 10$
 $+ = 2$
 $- = 8 \Rightarrow X = 2$

$$H_0: p = 0.5, H_a: p \neq 0.5$$

$$X \sim \text{Bin}(10, 0.5)$$

$$P(X=0) = \binom{10}{0} 0.5^{10} = 0.0010$$

$$P(X=1) = \binom{10}{1} 0.5^{10} = 0.0098$$

$$P(X=2) = \binom{10}{2} 0.5^{10} = 0.0439$$

$$2 \times (0.0010 + 0.0098) = 0.0216$$

$$\Rightarrow R.R = X \leq 1 \text{ or } X \geq 9$$

$$\Rightarrow X = 2 \text{ 不落入 } R.R.$$

$$\Rightarrow \text{Non-Reject } H_0$$

b. 資料非連續常態變數，各家醫院樣本數差異極大，導致變異數不同